

Flow Matching

NASA AI/ML STIG
Tomasz Różański
RSAA@ANU

Where this talk sits in the STIG schedule?

Module 5: Generative Models:

- Mar 16 - Normalizing Flows - Gregory Green
 - Normalizing Flows, Continuous Normalizing Flows (CNF), conditional models
- Mar 23 - Diffusion Models - Duo Xu
 - Denoising Diffusion Probabilistic Models, efficient sampling, model as a learned prior
- **Mar 30 - Flow Matching** - Tomasz Róžański
- Apr 6 - Simulation Based Inference (SBI) - Tomasz Róžański

Why astronomers care about distributions?

$$p(x)$$

- Spectra, images, time series
- Abundances
- Initial Mass Function
- CMD
- ...

$$p(x \mid \theta)$$

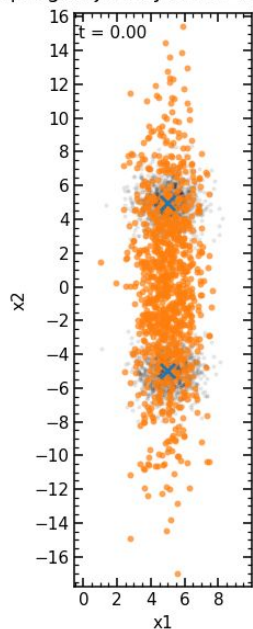
Forward
population/simulator
models

$$p(\theta \mid x)$$

Parameter inference /
SBI

One unifying picture: a probability path

Sampling trajectory of the learned flow



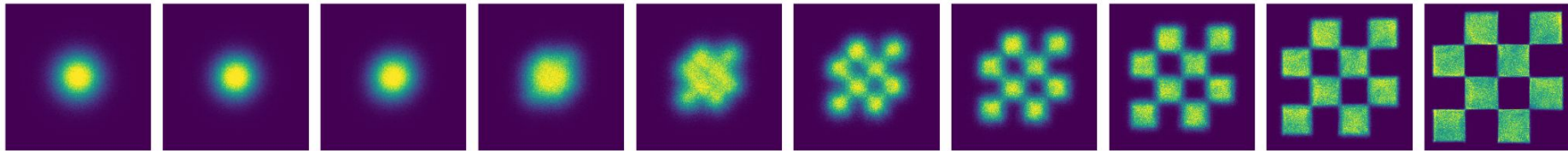
$$\dot{x}_t = u_t(x_t)$$

$$\partial_t p_t(x) = -\nabla \cdot (p_t(x_t) u_t(x_t))$$

CNF: the learned object and how the sampling works?

$$\frac{d}{dt}\psi_t(x_0) = u(\psi_t(x_0), t) \quad x_t = \psi_t(x_0)$$

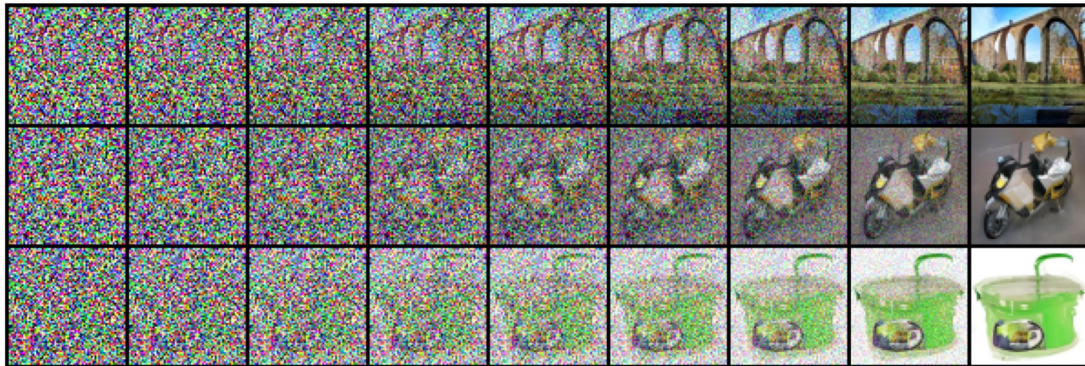
$$\log p_1(x_1) = \log p_0(x_0) - \int_0^1 \nabla \cdot u(x_t, t) dt$$



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How to train a model to regress this velocity field?

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Maximum likelihood training?

$$\hat{\phi}_{\text{MLE}} = \arg \max_{\phi} \sum_{i=1}^N \log p_{1,\phi}(x_i)$$

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How to train a model to regress this velocity field?

Maximum likelihood training?

$$\hat{\phi}_{\text{MLE}} = \arg \max_{\phi} \sum_{i=1}^N \log p_{1,\phi}(x_i) = \arg \max_{\phi} \sum_{i=1}^N \left(\log p_0(x_{0,i}) - \int_0^1 \nabla \cdot u_{\phi}(x_{t,i}, t) dt \right)$$

Flow matching: simulation-free training for CNFs

$$\mathcal{L}_{\text{FM}}(\phi) = \mathbb{E}_{t \sim U[0,1], x \sim p_t} [\|u_\phi(x, t) - u(x, t)\|^2]$$



Flow matching: simulation-free training for CNFs

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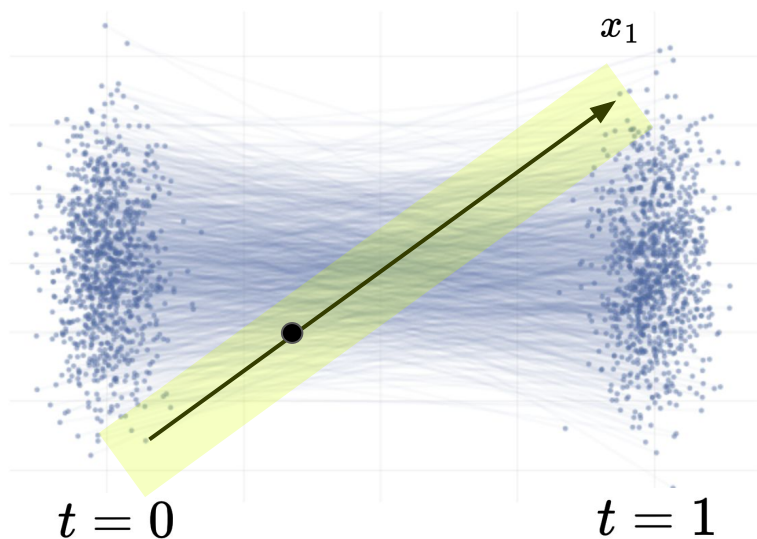
Conditional Flow Matching to the rescue!

$$\mathcal{L}_{\text{FM}}(\phi) = \mathbb{E}_{t \sim U[0,1], x \sim p_t} [\|u_\phi(x, t) - u(x, t)\|^2]$$

$$\mathcal{L}_{\text{CFM}}(\phi) = \mathbb{E}_{t \sim U[0,1], x_1 \sim q, x_t \sim p_t(x|x_1)} [\|u_\phi(x, t) - u_t(x|x_1)\|^2]$$

$$\nabla_\phi \mathcal{L}_{\text{FM}} = \nabla_\phi \mathcal{L}_{\text{CFM}}$$

Pure Gaussian worked example



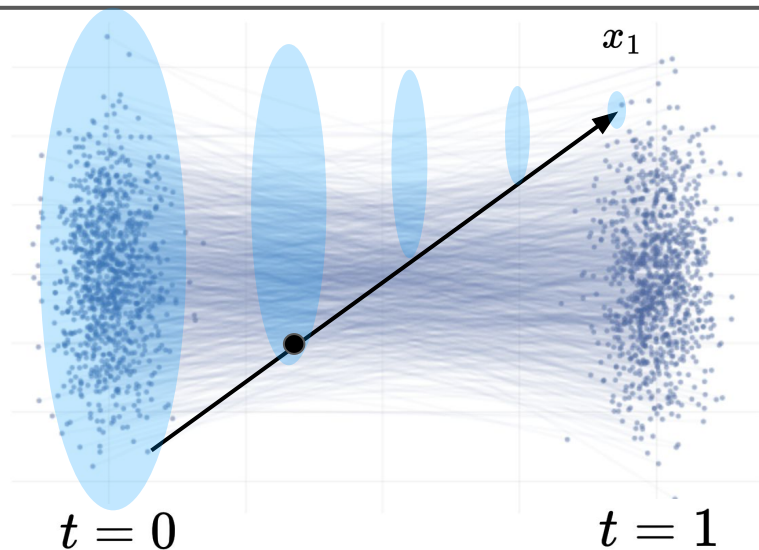
$$x_t = (1 - t)x_0 + tx_1$$

$$p_t(x|x_1) \sim \mathcal{N}(tx_1, (1 - t)^2 I)$$

$$\dot{x}_t = u_t(x|x_1) = \frac{x_1 - x}{1 - t}$$

$$\mathcal{L}_{\text{CFM}}(\phi) = \mathbb{E}_{t \sim U[0,1], x_1 \sim q, x_t \sim p_t(x|x_1)} [\|u_\phi(x, t) - u_t(x|x_1)\|^2]$$

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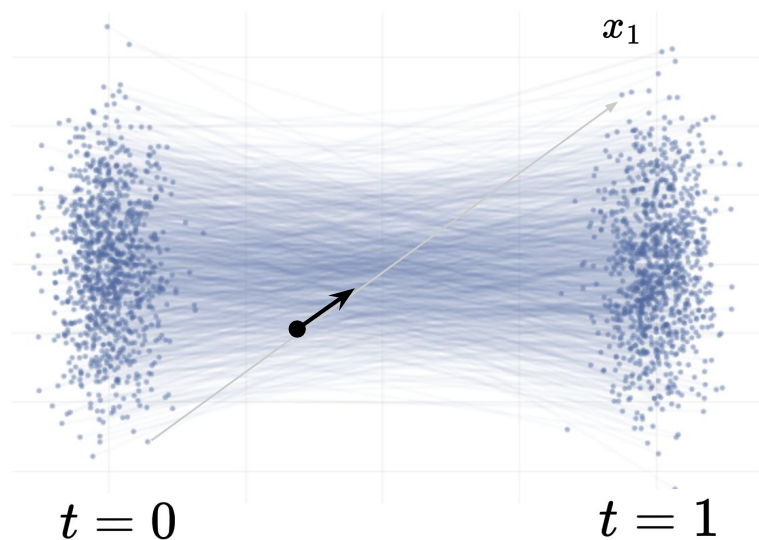
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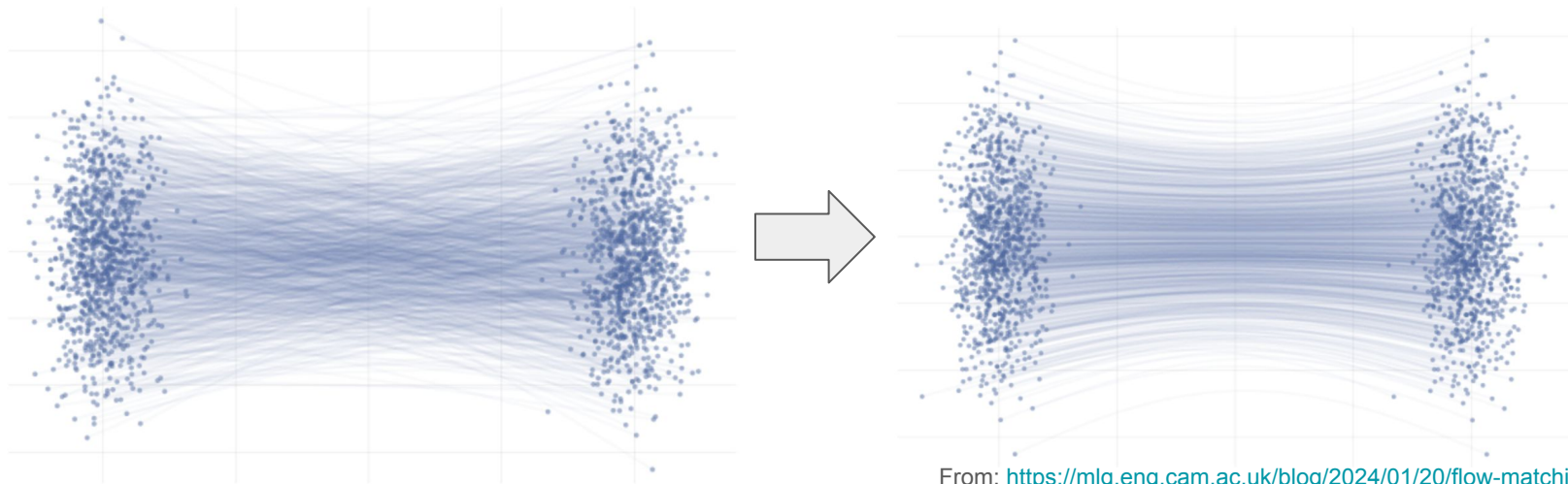
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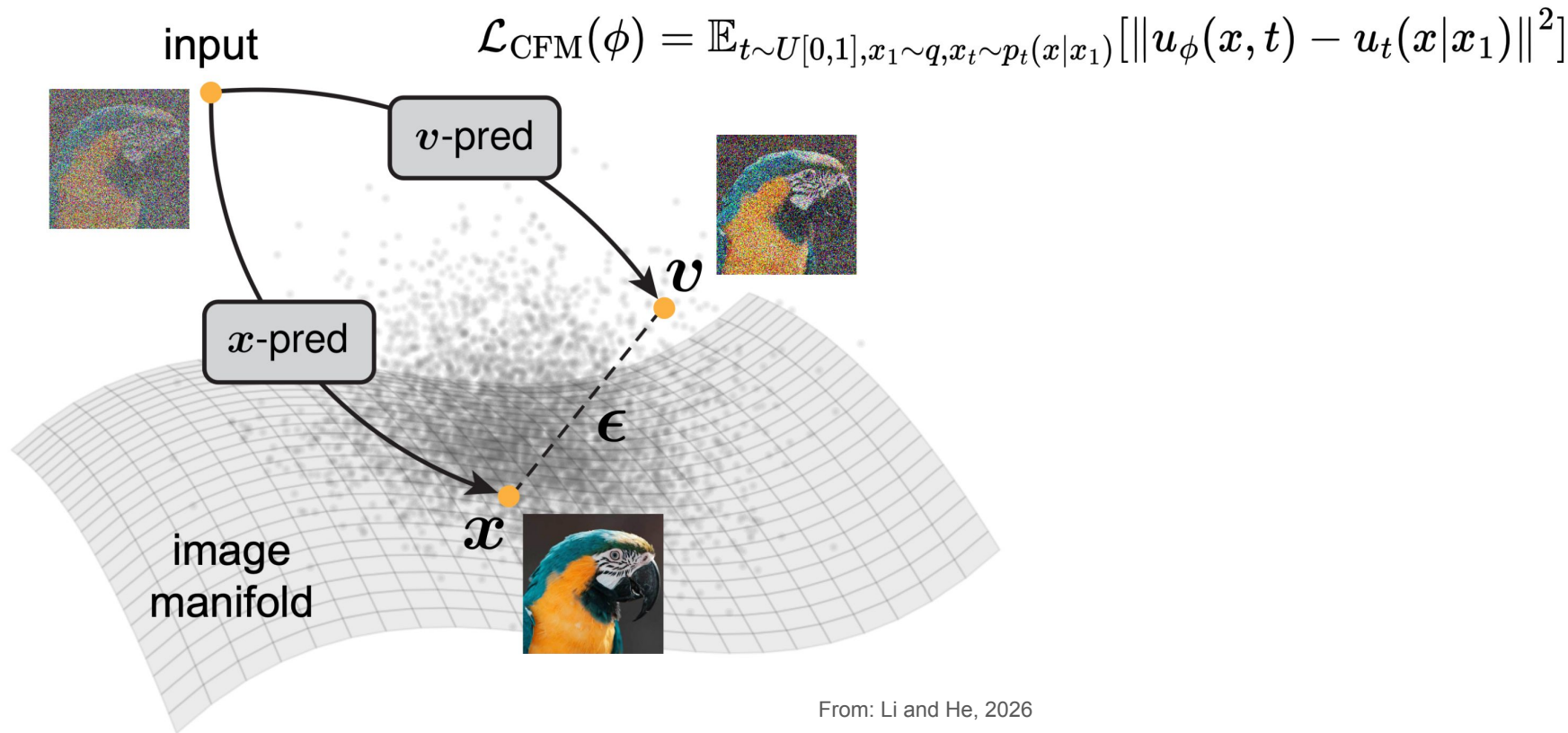
Conditional Flow Matching to the rescue!

$$\mathcal{L}_{\text{FM}}(\phi) = \mathbb{E}_{t \sim U[0,1], x \sim p_t} [\|u_\phi(x, t) - u(x, t)\|^2]$$

$$\mathcal{L}_{\text{CFM}}(\phi) = \mathbb{E}_{t \sim U[0,1], x_1 \sim q, x_t \sim p_t(x|x_1)} [\|u_\phi(x, t) - u_t(x|x_1)\|^2]$$



What does the network predict? What it should predict?



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$$\mathcal{L}_{\text{CFM}}(\phi) = \mathbb{E}_{t \sim U[0,1], x_1 \sim q, x_t \sim p_t(x|x_1)} [\|u_\phi(x, t) - u_t(x|x_1)\|^2]$$

	(a) x -pred $\mathbf{x}_\theta := \text{net}_\theta(\mathbf{z}_t, t)$	(b) ϵ -pred $\boldsymbol{\epsilon}_\theta := \text{net}_\theta(\mathbf{z}_t, t)$	(c) v -pred $\mathbf{v}_\theta := \text{net}_\theta(\mathbf{z}_t, t)$
(1) x -loss: $\mathbb{E} \ \mathbf{x}_\theta - \mathbf{x}\ ^2$	$\mathbf{x}_\theta$	$\mathbf{x}_\theta = (\mathbf{z}_t - (1-t)\boldsymbol{\epsilon}_\theta)/t$	$\mathbf{x}_\theta = (1-t)\mathbf{v}_\theta + \mathbf{z}_t$
(2) ϵ -loss: $\mathbb{E} \ \boldsymbol{\epsilon}_\theta - \boldsymbol{\epsilon}\ ^2$	$\boldsymbol{\epsilon}_\theta = (\mathbf{z}_t - t\mathbf{x}_\theta)/(1-t)$	$\boldsymbol{\epsilon}_\theta$	$\boldsymbol{\epsilon}_\theta = \mathbf{z}_t - t\mathbf{v}_\theta$
(3) v -loss: $\mathbb{E} \ \mathbf{v}_\theta - \mathbf{v}\ ^2$	$\mathbf{v}_\theta = (\mathbf{x}_\theta - \mathbf{z}_t)/(1-t)$	$\mathbf{v}_\theta = (\mathbf{z}_t - \boldsymbol{\epsilon}_\theta)/t$	$\mathbf{v}_\theta$

From: Li and He, 2026

What does the network predict?

$$\mathcal{L}_{\text{CFM}}(\phi) = \mathbb{E}_{t \sim U[0,1], x_1 \sim q, x_t \sim p_t(x|x_1)} [\|u_\phi(x, t) - u_t(x|x_1)\|^2]$$

	(a) x -pred	(b) ϵ -pred	(c) v -pred
	$x_\theta := \text{net}_\theta(z_t, t)$	$\epsilon_\theta := \text{net}_\theta(z_t, t)$	$v_\theta := \text{net}_\theta(z_t, t)$
(1) x -loss: $\mathbb{E}\ x_\theta - x\ ^2$	x_θ	$x_\theta = (z_t - (1-t)\epsilon_\theta)/t$	$x_\theta = (1-t)v_\theta + z_t$
(2) ϵ -loss: $\mathbb{E}\ \epsilon_\theta - \epsilon\ ^2$	$\epsilon_\theta = (z_t - t x_\theta)/(1-t)$	ϵ_θ	$\epsilon_\theta = z_t - t v_\theta$
(3) v -loss: $\mathbb{E}\ v_\theta - v\ ^2$	$v_\theta = (x_\theta - z_t)/(1-t)$	$v_\theta = (z_t - \epsilon_\theta)/t$	v_θ

Prediction: $u_\phi(x, t) = (x_\phi(x, t) - x_t)/(1-t)$

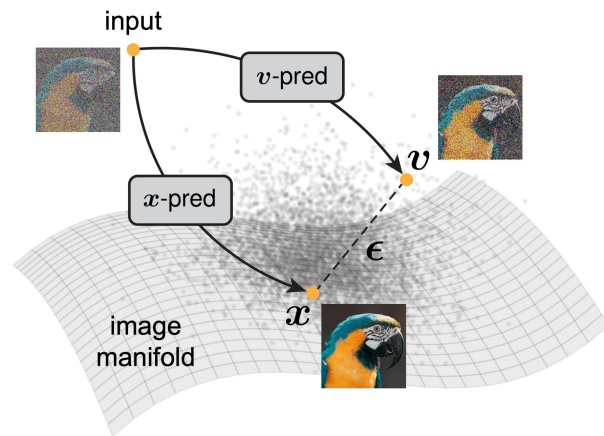
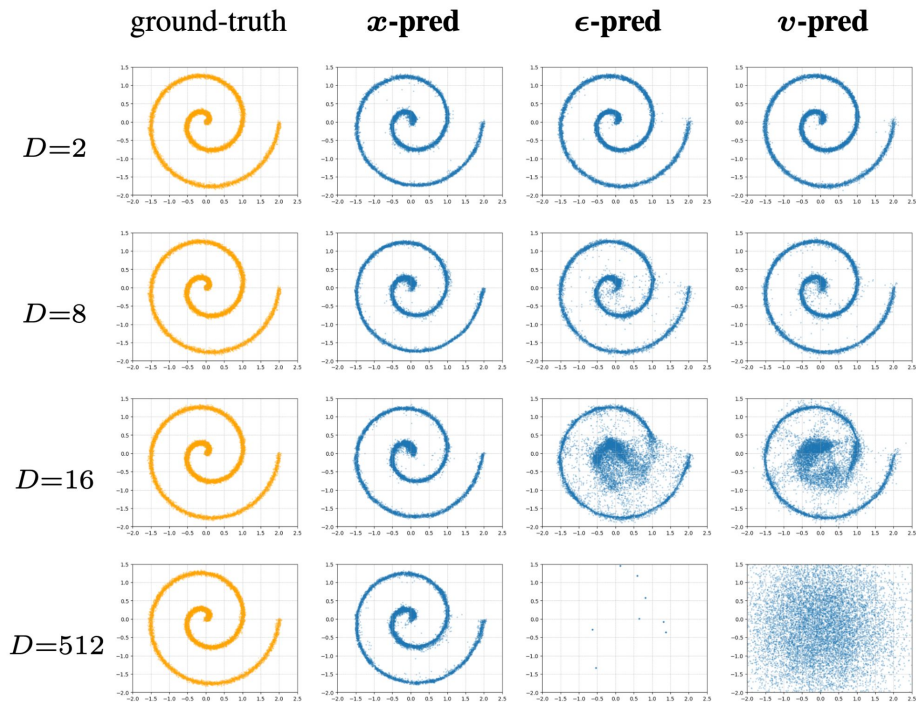
Target: $u(x, t) = (x_1 - x_t)/(1-t)$

$$\mathcal{L} = \mathbb{E}\|u_\phi(x, t) - u_t(x|x_1)\|^2 = \mathbb{E} \frac{1}{(1-t)^2} \|x_\phi(x, t) - x_1\|^2$$

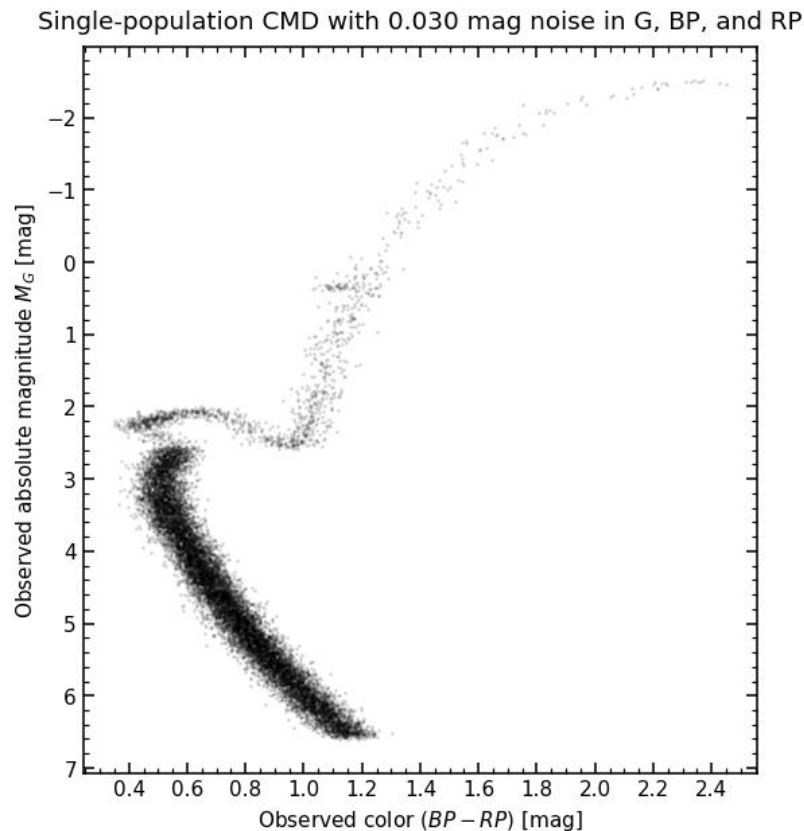
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What does the network predict? What it should predict?

$$\mathcal{L}_{\text{CFM}}(\phi) = \mathbb{E}_{t \sim U[0,1], x_1 \sim q, x_t \sim p_t(x|x_1)} [\|u_\phi(x, t) - u_t(x|x_1)\|^2]$$



Notebook 1 : modeling CMD of simple stellar population



Two meanings of conditional

- CFM conditioning
- Training-time endpoint conditioning

$$p_t(x|x_1)$$

- Scientific conditioning
- Modeling family of distributions

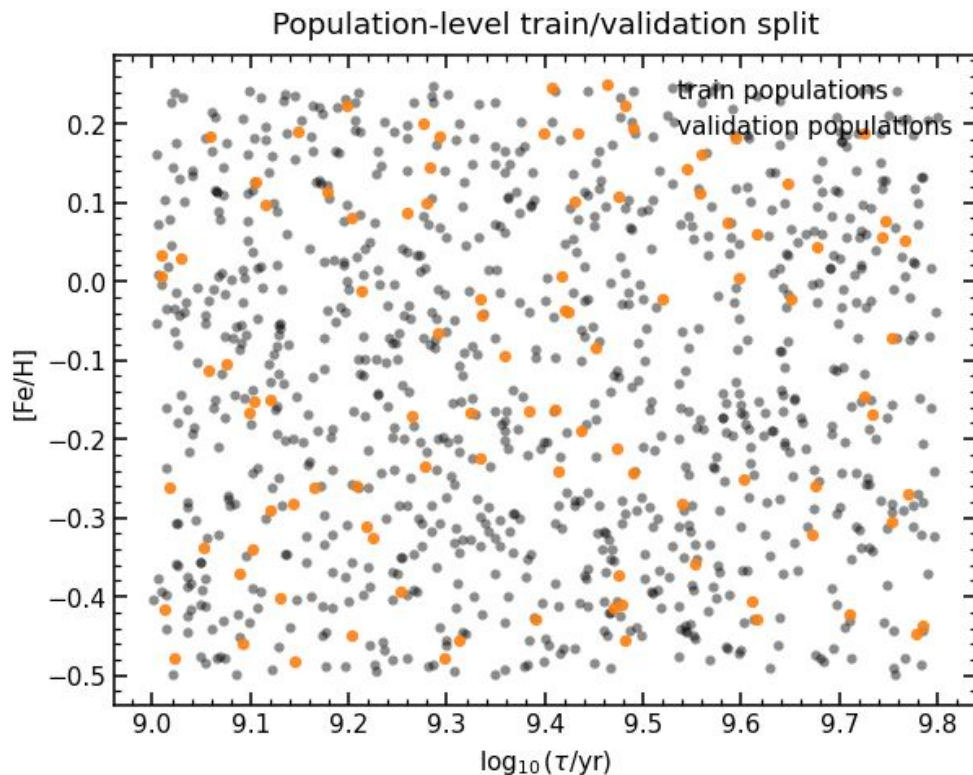
$$p(x|\theta)$$

Notebook 2 : modeling conditional distribution

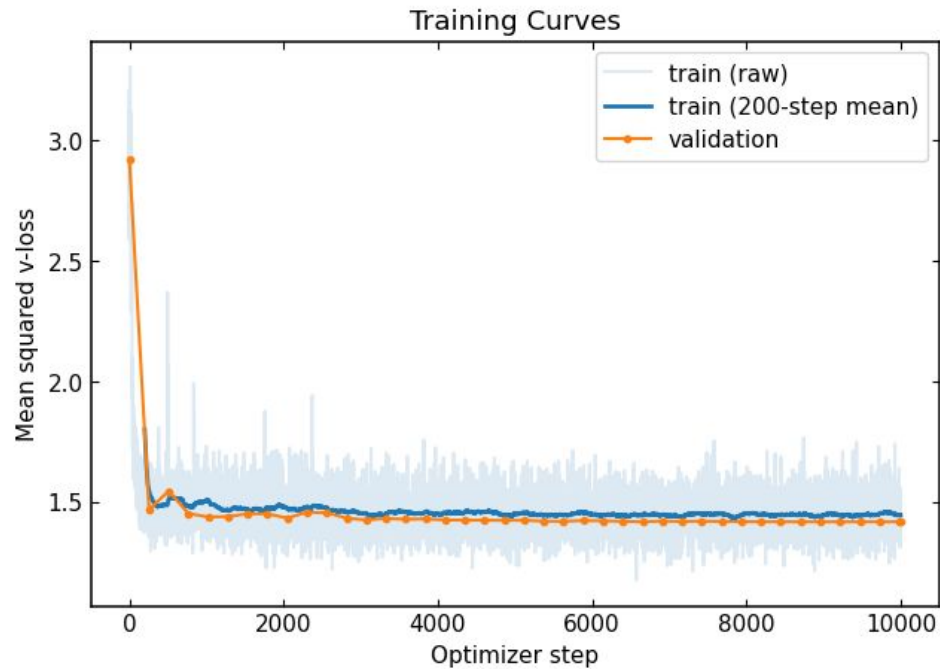
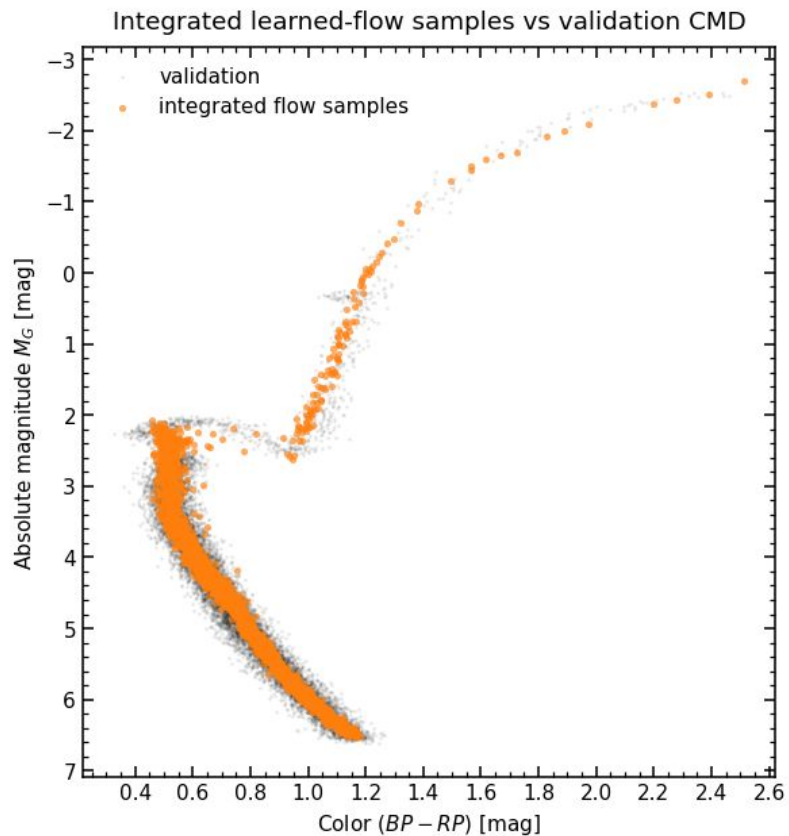
$$u_{\phi}(x, t) \rightarrow u_{\phi}(x, t, \theta)$$

$$p(x) \rightarrow p(x|\theta)$$

$$\theta = (\log \tau, [\text{Fe}/\text{H}])$$

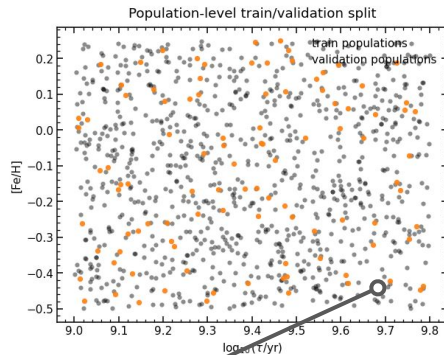
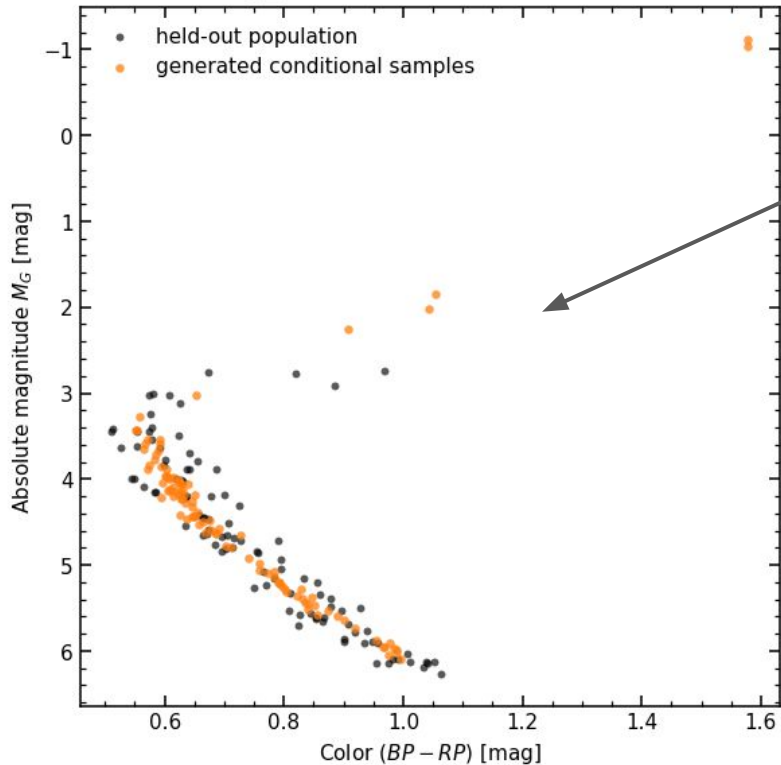


How to know it worked?

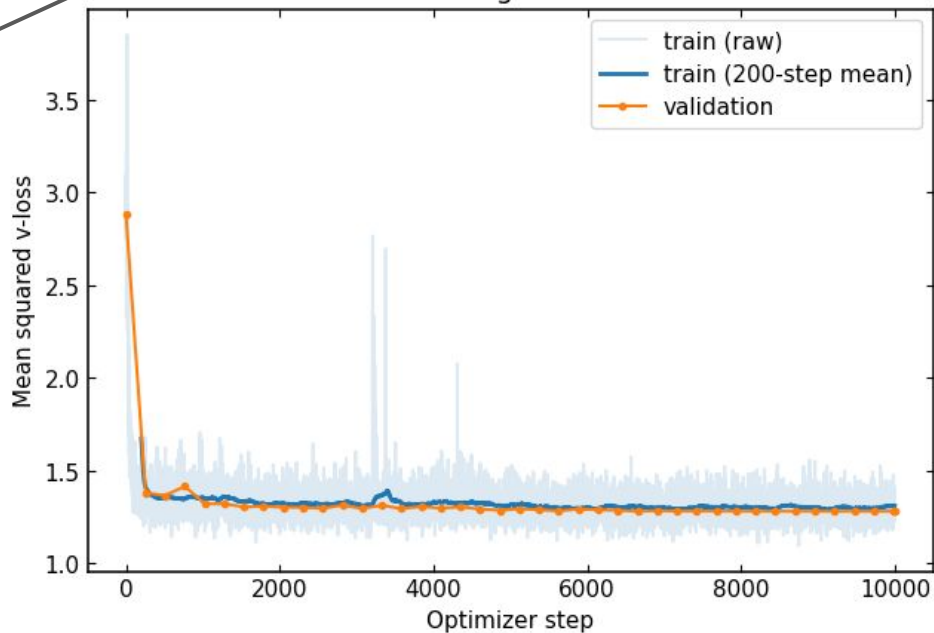


How to know it worked?

Held-out population pop_557
 $\log_{\text{tau}} = 9.711$, $[\text{Fe}/\text{H}] = -0.422$



Training Curves



Summary

- A unifying transport perspective links flows, diffusion, and variational methods.
- Flow Matching trains the vector field directly, avoiding ODE solves during training.
- Trainability depends strongly on what the network is asked to regress: velocity, noise, or data.
- Conditional flows are a practical and powerful framework for modelling conditional probabilities in astrophysics.
- Next step: turn forward generative models into reliable tools for simulation-based inference.